

SS Credit: Heat Island Reduction

This credit applies to

- ▶ BD+C: New Construction (1-4 points)
- ▶ BD+C: Core and Shell (1-4 points)
- ▶ BD+C: Schools (1-4 points)
- ▶ BD+C: Retail (1-4 points)
- ▶ BD+C: Data Centers (1-4 points)
- ▶ BD+C: Warehouses and Distribution Centers (1-4 points)
- ▶ BD+C: Hospitality (1-4 points)
- ▶ BD+C: Healthcare (1-4 points)

INTENT

To minimize inequitable effects on microclimates and human, especially frontline communities, and wildlife habitats by reducing heat islands.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Choose one of the following options:

Option 1. Nonroof and Roof (2 points except Healthcare, 1 point Healthcare)

Meet the following criterion:

$$\begin{array}{ccccccc} \text{Area of Nonroof Measures} & & \text{Area of High-Reflectance Roof} & & \text{Area of Vegetated Roof} & & \\ \text{-----} & + & \text{-----} & + & \text{-----} & \geq & \text{Total Site Paving Area} + \text{Total Roof Area} \\ 0.5 & & 0.75 & & 0.75 & & \end{array}$$

Alternatively, an SRI and SR weighted average approach may be used to calculate compliance.

Use any combination of the following strategies.

Nonroof Measures

- ▶ Use the existing plant material or install plants that provide shade over paving areas (including playgrounds) on the site within 10 years of planting. Install vegetated planters. Plants must be in place at the time of occupancy permit and cannot include artificial turf.
- ▶ Provide shade with structures covered by energy generation systems, such as solar thermal collectors, photovoltaics, and wind turbines.
- ▶ Provide shade with architectural devices or structures. If the device or structure is a roof, it shall have an aged *solar reflectance (SR)* value of at least 0.28 as measured in accordance with ANSI/CRRC S100. If the device or structure is not a roof, or if aged solar reflectance information is not available, it shall have at installation an initial SR of at least 0.33 as measured in accordance with ANSI/CRRC S100.
- ▶ Provide shade with vegetated structures.
- ▶ Use paving materials with an initial *solar reflectance (SR)* value of at least 0.33.
- ▶ Use an *open-grid pavement system* (at least 50% unbound).

High-Reflectance Roof

Use roofing materials that have an aged SRI equal to or greater than the values in Table 1. If aged SRI is not available, the roofing material shall have an initial SRI equal to or greater than the values in Table 1.

Table 1. Minimum solar reflectance index value, by roof slope

	Slope	Initial SRI	Aged SRI
Low-sloped roof	≤ 2:12	82	64
Steep-sloped roof	> 2:12	39	32

Roof area that consists of functional, usable spaces (such as helipads, recreation courts, and similar amenity areas) may meet the requirements of nonroof measures. Applicable roof area excludes roof area covered by mechanical equipment, solar energy panels, skylights, and any other appurtenances.

Vegetated Roof

Install a vegetated roof using native or adapted plant species.

OR

Option 2. Parking under Cover (1 point)

Place a minimum of 75% of *parking spaces under cover*. Any roof used to shade or cover parking must (1) have an aged SRI of at least 32 (if aged value information is not available, use materials with an initial SRI of at least 39 at installation), (2) be a vegetated roof, or (3) be covered by energy generation systems, such as solar thermal collectors, photovoltaics, and wind turbines.

The credit calculations must include all existing and new off-street parking spaces that are leased or owned by the project, including parking that is outside the project boundary but is used by the project. On-street parking in public rights-of-way is excluded from these calculations.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

This updated credit references the new ANSI/Cool Roof Rating Council S100 standard for “Rapid Ratings.” Updates also clarify that functional space for occupant use located on a roof can be included under nonroof calculations. To ensure that this credit is maximizing its intended impact, projects pursuing Option 2. Parking Under Cover are required to include all off-street parking used by the project in their calculations.

Step-by-Step Guidance

Refer to the LEED v4 reference guide, with the following modification:

- ▶ Identify hardscape and roof area
 - Common roads that serve multiple buildings should not be included in this calculation. If the roads include the primary parking areas (i.e. parking spots along the road), those parking areas should be included in the calculation.
- ▶ Selecting roofing materials:
 - Initial SRI shall be computed in accordance with ASTM E1980, under medium wind-speed conditions from initial values of solar reflectance and thermal emittance measured in accordance with ANSI/CRRC S100.
 - Aged SRI shall be computed in accordance with ASTM E1980, under medium wind-speed conditions from aged values of solar reflectance and thermal emittance measured in accordance with ANSI/CRRC S100.
 - Values of roof solar reflectance, thermal emittance, and solar reflectance index may be available at the Cool Roof Ratings Council's Rated Products Directory.
- ▶ Evaluate compliance against credit requirements:

- Equation 2, within the “summed for all high-reflectance roof areas” portion, replace ‘Required SR’ in the denominator with ‘Required SRI’.
- Credit calculations must include all existing and new off-street parking spaces owned or leased by the project.

Further Explanation

Referenced Standards

- ▶ ANSI/CRRRC S100, Standard Test Methods for Determining Radiative Properties of Material
- ▶ ASTM E1980 Standard Practice for Calculating Solar Reflectance Index of Horizontal and Low-Sloped Opaque Surfaces
- ▶ Cool Roof Ratings Council, Rated Products Directory: <https://coolroofs.org/directory>

Definitions

- ▶ **common road** for the purposes of this credit, a common road is defined as a narrow or small road or alleyway between buildings, which may or may not be drivable.
- ▶ **solar reflectance (SR)** The ratio of the reflected solar flux to the incident solar flux.
- ▶ **solar reflectance, initial (initial SR)** a solar reflectance that is measured prior to aging.
- ▶ **solar reflectance, aged (aged SR)** a solar reflectance that is measured after laboratory or natural aging.
- ▶ **solar reflectance index (SRI)** a measure of the constructed surface’s ability to stay cool in the sun by reflecting solar radiation and emitting thermal radiation. A standard black surface has an initial SRI of 0, and a standard white surface has an initial SRI of 100.
- ▶ **thermal emittance (TE)** the ratio of the radiant heat flux emitted by a specimen to that emitted by a blackbody radiator at the same temperature.
- ▶ **thermal emittance, initial (initial TE)** a thermal emittance that is measured prior to aging.
- ▶ **thermal emittance, aged (aged TE)** a thermal emittance that is measured after laboratory or natural aging.

Connection to Ongoing Performance:

- ▶ LEED O+M SS Credit Rainwater Management: Reducing a site’s hardscape area and/or using open-grid paving will improve infiltration rates and may help projects earn the related performance-based credit. Vegetated roofs can also contribute to rainwater management.